G20
SOUTH AFRICA 2025

G20 RESEARCH AND INNOVATION WORKING GROUP

MPUMALANGA ROADMAP: CONSIDERATIONS (OR PRINCIPLES) FOR A GLOBAL NATURAL HISTORY COLLECTION ROADMAP

PRIORITY 2: BIODIVERSITY INFORMATION FOR SUSTAINABLE DEVELOPMENT

DELIVERABLE 2.2 - NATURAL HISTORY COLLECTIONS COLLABORATION SEMINAR AND WORKSHOP

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Considerations (or Principles) for a Global Natural History Collection Roadmap

Based on the recommendations received during the G20 RIWG Seminar and Workshop: Developing Global Collaborations Supporting Museums and Natural History Collections.

Strategic frameworks, including implementation roadmaps, are most effective when shaped through broad consultation, inclusive participation, and strong community ownership and buy-in. The G20 Research and Innovation Working Group (RIWG) seminar represented an important and productive first step in bringing together diverse voices from across the natural history collections (NHC) community. Through the sharing of experiences, ideas, and ambitions, the seminar laid a strong foundation for future collaboration.

The insights gathered during this process helped to identify key principles, common goals, and shared priorities that can inform long-term strategic planning. The seminar catalysed the development of a baseline for a global roadmap by surfacing common goals, sparking momentum for collaboration, and setting the stage for deeper engagement across the NHC community, which will require adequate resourcing to produce a holistic and authoritative global roadmap. In 2025 this deliverable is for noting by the G20, and does not require/imply financial commitments from governments or their supported institutions in this current year.

The following considerations should be taken into account by the natural history community in the consolidation of a roadmap, to take place over a short, medium, or longer-term framework, to develop a successful collaboration among all global natural history collections.

1. Mutually Beneficial Collaboration
2. Optimizing the human knowledge embedded in natural history collections
3. Infrastructure improvements and integrations
4. The Future of Collections

Each thematic area could include specific goals that can be accomplished by Museums and Natural History Collections Institutions.

Unifying Vision/Overarching Goal

The overarching goal of the Global Natural History Collections Roadmap, as proposed by the G20 Seminar on Natural History Collections, is to enhance international cooperation related to natural history research infrastructures and build global science capacity to further develop them. A critical element of this goal is capacity development, ensuring that expertise, skills, and resources are strengthened equitably across regions. By agreeing on and applying common standards, aligning existing processes, and leveraging synergies, institutions worldwide can ensure that research infrastructures, data systems, and collections are interoperable, accessible, and positioned to address urgent global challenges such as biodiversity loss, climate change, and emerging diseases.

Focus Area 1: Mutually Beneficial Collaboration

Establish inclusive partnerships at national, regional, and international scales

Goal 1: Commit to FAIR open data and global collaboration among natural history collections worldwide.

Goal 2: Develop new and apply successful mechanisms to develop regional natural history networks, building on proven impactful communities.

Goal 3: Collaborate across all sectors/disciplines and interested parties, involve citizen science in biodiversity research, and especially include Indigenous People and their Knowledge Systems, and the CARE principles.

Focus Area 2: Optimize The Human Knowledge Embedded In Natural History Collections

Facilitate exchanges among natural history scientists

Goal 4: Develop a biodiversity informatics and geological informatics training clearing-house to increase human capacity and build technical skills in biodiversity informatics, benefiting from technological advances (integrating AI models, genomics, etc., to its full extent).

Goal 5: Create and fund a system of secondments and sabbaticals to facilitate in-person interactions and expertise mobilization.

Goal 6: Develop international working groups to address prioritized knowledge gaps (Training, Curation, Specimen Collection, and Collecting Prioritization, etc.).

Focus Area 3: Infrastructure Improvements and Integrations

Build upon technical successes to include all natural history collections

Goal 7: Understand the strengths and fill the gaps in current natural history collections, and align with international legal collection and exploitation permits.

Goal 8: Promote the use of TDWG data standards in collection data management systems to increase interoperability, enabling enhanced data analysis for research, decision-making, and socio-economic benefit.

Goal 9: Collaborate with allied communities such as research field stations, biobanking, genetic resources, and many more, following a multidisciplinary approach integrating life and social sciences to integrate and escalate knowledge and expertise (agro-food, health, fisheries, etc).

Focus Area 4: The Future of Collections

Implement the vision immediately

Goal 10: Describe the vision for natural history collections as an ever-expanding research infrastructure that includes ambitious and impactful collections strategies, in a new world of machine-derived data and Artificial Intelligence.

Goal 11: Clarify and amplify the message of the role of natural history collections in public policy, especially under the current environmental crisis, and support the production of actionable knowledge to secure sustainable growth, a resilient society, and robust response mechanisms in the science-policy interface (e.g., One Health linking life sciences, social sciences, and humanities, for the well-being of humans, animals, and ecosystems).

Goal 12: Engage with private sector data holders to broaden impact, and facilitate their embedment into biodiversity data use (as for complying with policies like the Corporate Sustainability Reporting Directive (CSRD) under the EU).

Appendix 1

Mpumalanga Roadmap for Advancing Global Collaboration in Natural History Collections

Phase 1: Laying the Foundation (0–2 years)

Objectives: Establish partnerships, baseline assessments, initiate pilots, and mobilize early funding.

Key Actions	Milestones
Establish global and regional partnerships (R1, R2)	Formalized MOUs and collaboration frameworks
Develop and publish the roadmap and action plans (R1)	Roadmap endorsed by partner institutions
Map African NHC infrastructure (R9)	Completion of baseline database
Launch pilot AI-powered digitisation projects (R19)	First pilots operational
Create international and transnational working groups (R4, R5)	Thematic working groups established
Identify and secure sustainable funding sources (R6)	Seed funding for priority areas secured
Set up an Indigenous Knowledge & Ethics Taskforce (R10)	Framework for ethical engagement developed

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Phase 2: Scaling and Integration (2–5 years)

Objectives: Scale up digitisation, regional hubs, and training programs; align policy and governance frameworks.

Key Actions	Milestones
Develop regional hubs and natural history infrastructure networks outside the Global North (R7, R8)	At least 3 regional hubs launched
Develop and implement an integrated biodiversity informatics curriculum and training (R12, R21, R22)	Training centres and online platforms are active
Implement data and specimen integration (R19, R20)	The DOI system adopted in partner institutions
Align institutional and national policies with international digital heritage frameworks (R3, R11, R14)	National-level digital policies, including the policy on sensitive species data, are updated
Expand Indigenous science training programmes (R22)	Cohorts trained and placed in institutions

Phase 3: Consolidation and Innovation (5–10 years)

Objectives: Embed NHCs in science-policy interfaces, expand innovations, and measure impact.

Key Actions	Milestones
Launch Global Biodiversity Data and Network Hub (R3)	Operational and interoperable platform online
Establish and sustain joint field research stations (R13)	Multi-country research stations producing outputs
Operationalize biobanks for research and commercial use, and ensure relevant legal frameworks are put in place first (R16)	Biobanks network integrated with NHCs
Mainstream NHC contributions in national and global biodiversity policies, including biosecurity (R17, R18)	Evidence of policy influence and impact
Implement coordinated monitoring of GBF targets using NHC data	Target-relevant reporting adopted

Cross-Cutting Themes

- Equity & Inclusion: Prioritize Global South participation and Indigenous knowledge integration.
- Interoperability & Standards: Harmonize global data infrastructure
- Sustainability: Build financial and institutional sustainability into all phases.
- Ethics & Trust: Ensure transparent and community-driven data governance, using proven initiatives.
- Capacity Building

*R – Refers to the Recommendations identified in The Mpumalanga Report (Table 1).